

Unified Communications Platform (Teams Replica)

MINIMUM VIABLE PRODUCT (MVP) SPECIFICATION

| PROJECT OVERVIEW

The Teams Replica is an all-in-one unified communications platform that synthesizes persistent organizational text chat, tiered channel structures, and real-time virtual calling into a single environment.

| MAIN OBJECTIVE

The main objective of this platform is to bridge asynchronous team discussions with synchronous audio/video meetings, enabling communication across distinct departments.

| USER ROLES

The system maps standard team hierarchy structures onto structural platform gates:

- **Global Administrators:** Structural workspace setup, usage matrix checking, and permission tracking.
- **Team Owners:** Channel initialization, participant invitation curation, and board control.
- **Channel Members:** Group chat participation, call answering, and content distribution capabilities.

| CORE MVP FEATURES

Structured Teams & Channels

Enables users to create public or private organizational teams structured around departments, containing specialized topical channels.

Chat Engine (Personal & Group)

Provides instant one-to-one and group messaging complete with real-time typing indicators, read receipts, media/file attachments, and emoji reactions.

Integrated Video & Audio Calling

Enables immediate click-to-call functionality for direct peer-to-peer audio calls or scheduled multi-party video conferencing with integrated screen sharing.

File Repository & Sharing

Centralizes document sharing inside chat threads and channels, ensuring files are cataloged, searchable, and easily downloadable.

Activity Feed & Presence

Keeps users aligned through an aggregate notification timeline for direct mentions, combined with live presence states (Online, Busy, In a Meeting, DND).

| TECHNOLOGY STACK

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Additional Tools:** WebRTC (Calls/Video), Socket.IO (Chat & Presence), AWS S3

| CONCLUSION

The Teams Replica MVP offers an immediate communications core, eliminating fractured messaging and providing teams with a robust, real-time workspace for daily operations.